

Chapter 2

Vectors



Chapter 2 Vectors

Vectors in Two Dimensions

A vector is a quantity with both **magnitude** and **direction**. A vector in the coordinate plane is expressed in terms of two coordinates or components

$$\mathbf{v} = \langle a_1, a_2 \rangle$$

If a vector $\mathbf{v}$ has its initial point at $P(x_1, y_1)$ and its terminal point at $Q(x_2, y_2)$, then

$$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1 \rangle = \overrightarrow{PQ}$$

Let $\mathbf{u} = \langle a_1, a_2 \rangle$, $\mathbf{v} = \langle b_1, b_2 \rangle$, and $c \in \mathbb{R}$. The operations on vectors are defined as follows.

$$\mathbf{u} + \mathbf{v} = \langle a_1 + b_1, a_2 + b_2 \rangle$$

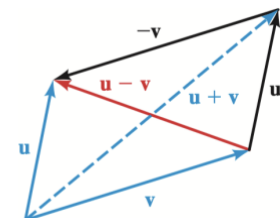
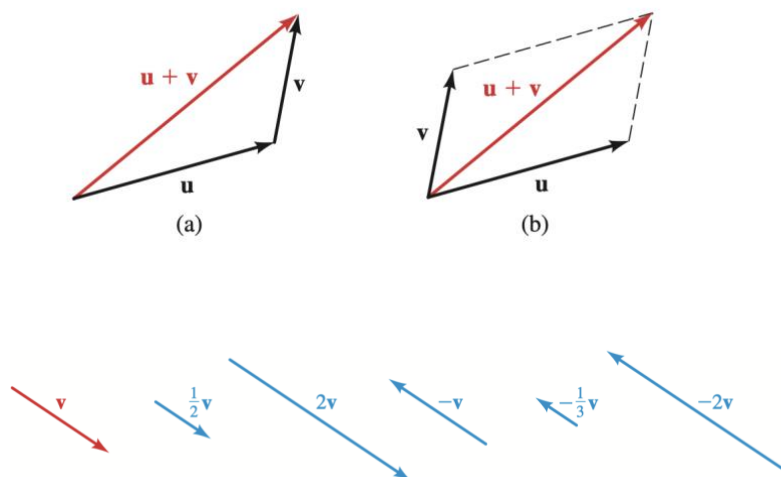
Addition

$$\mathbf{u} - \mathbf{v} = \langle a_1 - b_1, a_2 - b_2 \rangle$$

Subtraction

$$c\mathbf{u} = \langle ca_1, ca_2 \rangle$$

Scalar multiplication

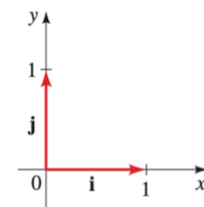


The vectors $\mathbf{i}$ and $\mathbf{j}$ are defined by

$$\mathbf{i} = \langle 1, 0 \rangle \quad \mathbf{j} = \langle 0, 1 \rangle$$

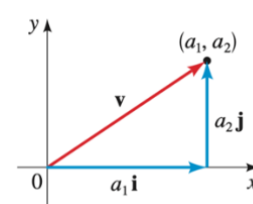
Any vector $\mathbf{v} = \langle a_1, a_2 \rangle$ can be expressed as

$$\mathbf{v} = a_1 \mathbf{i} + a_2 \mathbf{j}$$



Let $\mathbf{v} = \langle a_1, a_2 \rangle$. The magnitude (or length) of $\mathbf{v}$ is

$$|\mathbf{v}| = \sqrt{a_1^2 + a_2^2}$$



EXAMPLE 1 Magnitudes of Vectors

Find the magnitude of each vector.

(a) $\mathbf{u} = \langle 2, -3 \rangle$

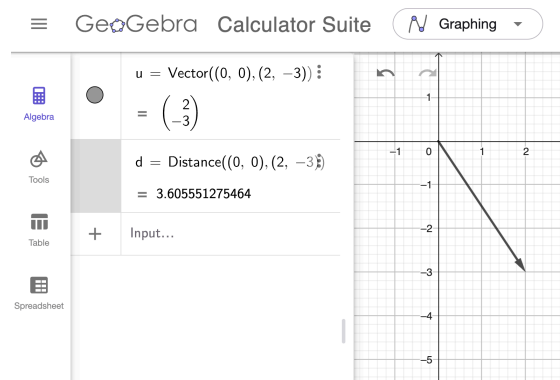
(b) $\mathbf{v} = \langle 5, 0 \rangle$

(c) $\mathbf{w} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$

SOLUTION

(a) $|\mathbf{u}| = \sqrt{2^2 + (-3)^2} = \sqrt{13}$

(b) $|\mathbf{v}| = \sqrt{5^2 + 0^2} = \sqrt{25} = 5$



$$(c) |\mathbf{w}| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(\frac{4}{5}\right)^2} = \sqrt{\frac{9}{25} + \frac{16}{25}} = \sqrt{\frac{25}{25}} = 1$$

A vector of length 1 is called a **unit vector**. For instance, in Example 1(c) the vector $\mathbf{w} = \left\langle \frac{3}{5}, \frac{4}{5} \right\rangle$ is a unit vector.

EXAMPLE 2 Operations with VectorsIf $\mathbf{u} = \langle 2, -3 \rangle$ and $\mathbf{v} = \langle -1, 2 \rangle$, find $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, $2\mathbf{u}$, $-3\mathbf{v}$, and $2\mathbf{u} + 3\mathbf{v}$.**SOLUTION** By the definitions of the vector operations we have

$$\mathbf{u} + \mathbf{v} = \langle 2, -3 \rangle + \langle -1, 2 \rangle = \langle 1, -1 \rangle$$

$$\mathbf{u} - \mathbf{v} = \langle 2, -3 \rangle - \langle -1, 2 \rangle = \langle 3, -5 \rangle$$

$$\begin{aligned} 2\mathbf{u} &= 2\langle 2, -3 \rangle = \langle 4, -6 \rangle \\ -3\mathbf{v} &= -3\langle -1, 2 \rangle = \langle 3, -6 \rangle \\ 2\mathbf{u} + 3\mathbf{v} &= 2\langle 2, -3 \rangle + 3\langle -1, 2 \rangle \\ &= \langle 4, -6 \rangle + \langle -3, 6 \rangle \\ &= \langle 1, 0 \rangle \end{aligned}$$

PROPERTIES OF VECTORS

Vector addition

$$\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$$

$$\mathbf{u} + (\mathbf{v} + \mathbf{w}) = (\mathbf{u} + \mathbf{v}) + \mathbf{w}$$

$$\mathbf{u} + \mathbf{0} = \mathbf{u}$$

$$\mathbf{u} + (-\mathbf{u}) = \mathbf{0}$$

Length of a vector

$$|c\mathbf{u}| = |c||\mathbf{u}|$$

Multiplication by a scalar

$$\begin{aligned}
c(\mathbf{u} + \mathbf{v}) &= c\mathbf{u} + c\mathbf{v} \\
(c + d)\mathbf{u} &= c\mathbf{u} + d\mathbf{u} \\
(cd)\mathbf{u} &= c(d\mathbf{u}) = d(c\mathbf{u}) \\
1\mathbf{u} &= \mathbf{u} \\
0\mathbf{u} &= \mathbf{0} \\
c\mathbf{0} &= \mathbf{0}
\end{aligned}$$

EXAMPLE 3 Vectors in Terms of $\mathbf{i}$ and $\mathbf{j}$

- (a) Write the vector $\mathbf{u} = \langle 5, -8 \rangle$ in terms of $\mathbf{i}$ and $\mathbf{j}$.
 (b) If $\mathbf{u} = 3\mathbf{i} + 2\mathbf{j}$ and $\mathbf{v} = -\mathbf{i} + 6\mathbf{j}$, write $2\mathbf{u} + 5\mathbf{v}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.

SOLUTION

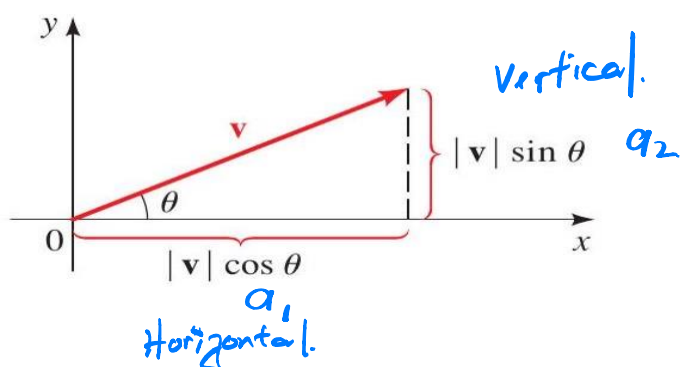
(a) $\mathbf{u} = 5\mathbf{i} + (-8)\mathbf{j} = 5\mathbf{i} - 8\mathbf{j}$

$$\begin{aligned}
(b) \quad 2\mathbf{u} + 5\mathbf{v} &= 2(3\mathbf{i} + 2\mathbf{j}) + 5(-\mathbf{i} + 6\mathbf{j}) \\
&= 6\mathbf{i} + 4\mathbf{j} - 5\mathbf{i} + 30\mathbf{j} \\
&= \mathbf{i} + 34\mathbf{j}
\end{aligned}$$

The direction of $\mathbf{v}$ is the smallest positive angle θ in standard position formed by the positive x -axis and $\mathbf{v}$ (see the figure below).

If $\mathbf{v} = \langle a_1, a_2 \rangle$, then the components of $\mathbf{v}$ satisfy

$$a_1 = |\mathbf{v}| \cos \theta \quad a_2 = |\mathbf{v}| \sin \theta$$



$$\begin{aligned}
\sin \theta &= \frac{a_2}{|\mathbf{v}|} \\
a_2 &= |\mathbf{v}| \sin \theta \\
\cos \theta &= \frac{a_1}{|\mathbf{v}|} \\
a_1 &= |\mathbf{v}| \cos \theta
\end{aligned}$$

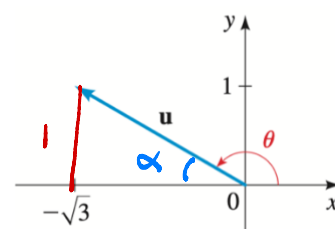
EXAMPLE 4 Components and Direction of a Vector

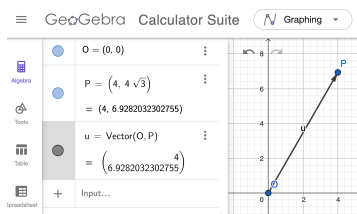
- (a) A vector $\mathbf{v}$ has length 8 and direction $\pi/3$. Find the horizontal and vertical components, and write $\mathbf{v}$ in terms of $\mathbf{i}$ and $\mathbf{j}$.
 (b) Find the direction of the vector $\mathbf{u} = -\sqrt{3}\mathbf{i} + \mathbf{j}$.

SOLUTION

(a) We have $\mathbf{v} = \langle a, b \rangle$, where the components are given by

$$\begin{aligned}
|\mathbf{v}| &= 8 \quad \theta = \frac{\pi}{3} \text{ rad} \\
a &= 8 \cos \frac{\pi}{3} = 4 \quad \text{and} \quad b = 8 \sin \frac{\pi}{3} = 4\sqrt{3} \\
\theta &= \frac{180^\circ}{3} = 60^\circ
\end{aligned}$$





Thus $\mathbf{v} = \langle 4, 4\sqrt{3} \rangle = 4\mathbf{i} + 4\sqrt{3}\mathbf{j}$.

(b) From Figure 15 we see that the direction θ has the property that

$$\tan \theta = \frac{1}{-\sqrt{3}} = -\frac{\sqrt{3}}{3}$$

$$\tan \alpha = \frac{1}{\sqrt{3}} \\ \alpha = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right)$$

Thus the reference angle for θ is $\pi/6$. Since the terminal point of the vector $\mathbf{u}$ is in Quadrant II, it follows that $\theta = 5\pi/6$.

$$\theta = \pi - \pi/6 = 5\pi/6 \text{ rad} = 30^\circ = \pi/6 \text{ rad} \\ \theta = 180^\circ - 30^\circ = 150^\circ \text{ degree.}$$

The Dot Product of Vectors

If $\mathbf{u} = \langle a_1, a_2 \rangle$ and $\mathbf{v} = \langle b_1, b_2 \rangle$, then their dot product is

$$\mathbf{u} \cdot \mathbf{v} = a_1b_1 + a_2b_2$$

If θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}||\mathbf{v}|\cos \theta$$

The angle θ between $\mathbf{u}$ and $\mathbf{v}$ satisfies

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}||\mathbf{v}|}$$

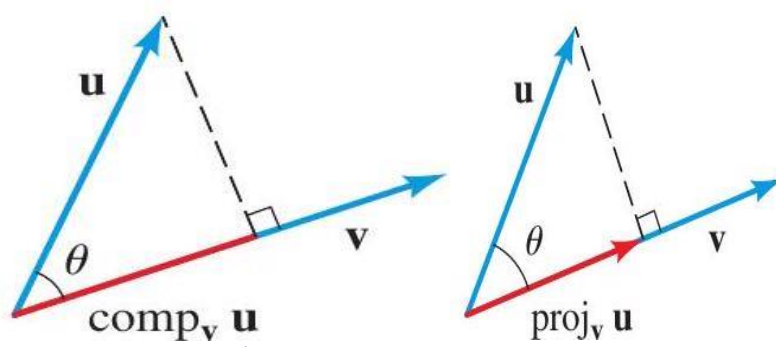
The vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if

$$\mathbf{u} \cdot \mathbf{v} = 0$$

$$\cos \theta = 0 \Leftrightarrow \theta = 90^\circ$$

The component of $\mathbf{u}$ along $\mathbf{v}$ (a scalar) and the projection of $\mathbf{u}$ onto $\mathbf{v}$ (a vector) are given by

$$\text{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|}, \quad \text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v}$$



EXAMPLE 5 Calculating Dot Products [5a,11a]

(a) If $\mathbf{u} = \langle 3, -2 \rangle$ and $\mathbf{v} = \langle 4, 5 \rangle$ then

$$\mathbf{u} \cdot \mathbf{v} = (3)(4) + (-2)(5) = 2$$

(b) If $\mathbf{u} = 2\mathbf{i} + \mathbf{j}$ and $\mathbf{v} = 5\mathbf{i} - 6\mathbf{j}$, then

$$\mathbf{u} \cdot \mathbf{v} = (2\mathbf{i} + \mathbf{j}) \cdot (5\mathbf{j} - 6\mathbf{j})$$

$$\mathbf{u} \cdot \mathbf{v} = (2)(5) + (1)(-6) = 4$$

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|}$$

EXAMPLE 6 Finding the Angle Between Two Vectors

Find the angle between the vectors $\mathbf{u} = \langle 2, 5 \rangle$ and $\mathbf{v} = \langle 4, -3 \rangle$.

SOLUTION

$$\cos \theta = \frac{(2)(4) + (5)(-3)}{\sqrt{2^2 + 5^2} \sqrt{4^2 + (-3)^2}} = \frac{-7}{5\sqrt{29}}$$

$$\theta = \cos^{-1}\left(\frac{-7}{5\sqrt{29}}\right) \approx 105.1^\circ$$

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are called **perpendicular**, or **orthogonal**, if the angle between them is $\pi/2$. The following theorem shows that we can determine whether two vectors are perpendicular by finding their dot product.

ORTHOGONAL VECTORS

Two nonzero vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if $\mathbf{u} \cdot \mathbf{v} = 0$.

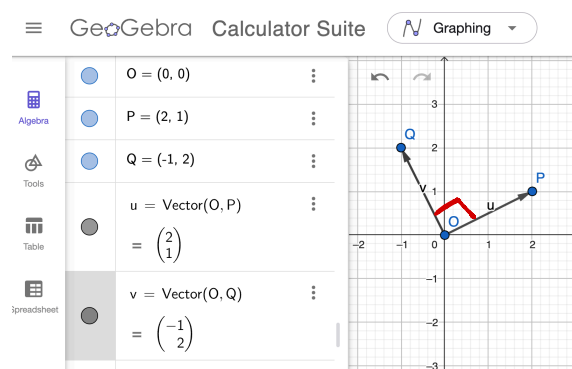
Proof. If $\mathbf{u}$ and $\mathbf{v}$ are perpendicular, then the angle between them is $\pi/2$, so

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \frac{\pi}{2} = 0$$

Conversely, if $\mathbf{u} \cdot \mathbf{v} = 0$, then

$$|\mathbf{u}| |\mathbf{v}| \cos \theta = 0$$

Since $\mathbf{u}$ and $\mathbf{v}$ are nonzero vectors, we conclude that $\cos \theta = 0$, so $\theta = \pi/2$. Thus $\mathbf{u}$ and $\mathbf{v}$ are orthogonal.



EXAMPLE 7 Checking Whether Two Vectors Are Perpendicular

Determine whether the vectors in each pair are perpendicular.

(a) $\mathbf{u} = \langle 3, 5 \rangle$ and $\mathbf{v} = \langle 2, -8 \rangle$

(b) $\mathbf{u} = \langle 2, 1 \rangle$ and $\mathbf{v} = \langle -1, 2 \rangle$

SOLUTION

(a) $\mathbf{u} \cdot \mathbf{v} = (3)(2) + (5)(-8) = -34 \neq 0$, so $\mathbf{u}$ and $\mathbf{v}$ are not perpendicular.

(b) $\mathbf{u} \cdot \mathbf{v} = (2)(-1) + (1)(2) = 0$, so $\mathbf{u}$ and $\mathbf{v}$ are perpendicular.

EXAMPLE 8 Finding Components

Let $\mathbf{u} = \langle 1, 4 \rangle$ and $\mathbf{v} = \langle -2, 1 \rangle$. Find the component of $\mathbf{u}$ along $\mathbf{v}$.

SOLUTION From the formula for the component of $\mathbf{u}$ along $\mathbf{v}$ we have

$$\text{comp}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|} = \frac{(1)(-2) + (4)(1)}{\sqrt{4 + 1}} = \frac{2}{\sqrt{5}}$$

THE VECTOR PROJECTION OF $\mathbf{u}$ ONTO $\mathbf{v}$

The projection of $\mathbf{u}$ onto $\mathbf{v}$ is the vector $\text{proj}_{\mathbf{v}} \mathbf{u}$ given by

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v} \quad \checkmark$$

If the vector $\mathbf{u}$ is resolved into $\mathbf{u}_1$ and $\mathbf{u}_2$, where $\mathbf{u}_1$ is parallel to $\mathbf{v}$ and $\mathbf{u}_2$ is orthogonal to $\mathbf{v}$, then

$$\mathbf{u}_1 = \text{proj}_{\mathbf{v}} \mathbf{u} \quad \text{and} \quad \mathbf{u}_2 = \mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u}$$

EXAMPLE 9 Resolving a Vector into Orthogonal Vectors

Let $\mathbf{u} = \langle -2, 9 \rangle$ and $\mathbf{v} = \langle -1, 2 \rangle$.

(a) Find $\text{proj}_{\mathbf{v}} \mathbf{u}$.

(b) Resolve $\mathbf{u}$ into $\mathbf{u}_1$ and $\mathbf{u}_2$, where $\mathbf{u}_1$ is parallel to $\mathbf{v}$ and $\mathbf{u}_2$ is orthogonal to $\mathbf{v}$.

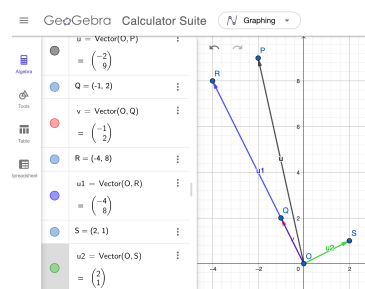
SOLUTION

(a) By the formula for the projection of one vector onto another we have

$$\begin{aligned} \text{proj}_{\mathbf{v}} \mathbf{u} &= \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{v}|^2} \right) \mathbf{v} && \text{Formula for projection} \\ &= \left(\frac{\langle -2, 9 \rangle \cdot \langle -1, 2 \rangle}{(-1)^2 + 2^2} \right) \langle -1, 2 \rangle \\ &= 4 \langle -1, 2 \rangle = \langle -4, 8 \rangle \quad \checkmark \end{aligned}$$

Formula for projection

(b) By the formula in the preceding box we have $\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2$, where

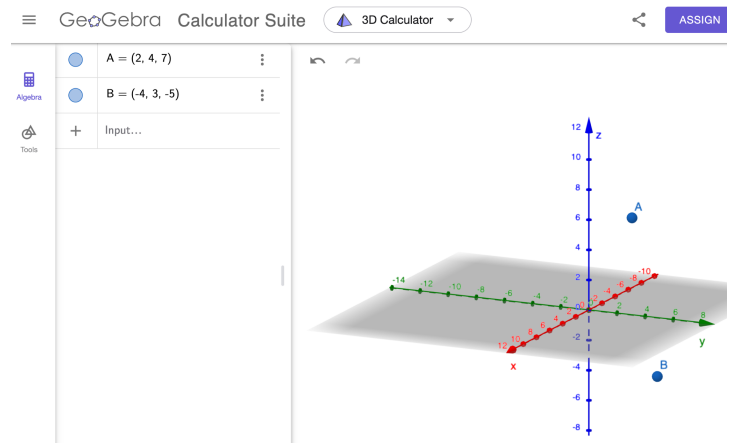
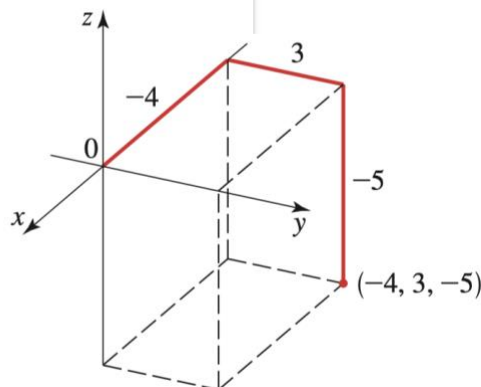
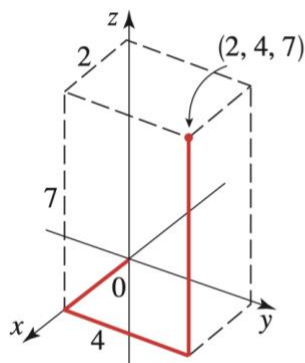
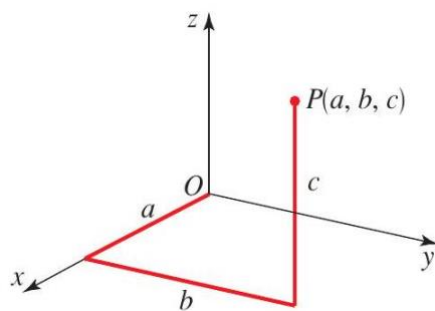


$$\mathbf{u}_1 = \text{proj}_{\mathbf{v}} \mathbf{u} = \langle -4, 8 \rangle$$

$$\mathbf{u}_2 = \mathbf{u} - \text{proj}_{\mathbf{v}} \mathbf{u} = \langle -2, 9 \rangle - \langle -4, 8 \rangle = \langle 2, 1 \rangle$$

Three-Dimensional Coordinate Geometry

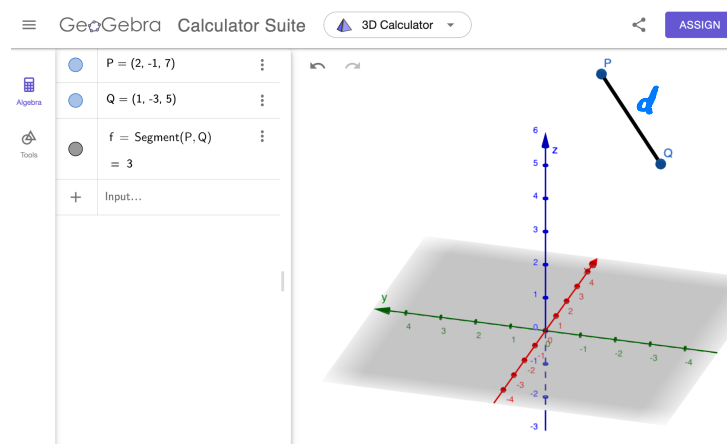
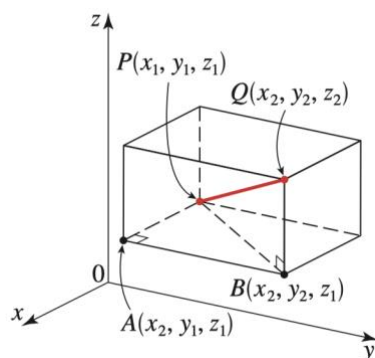
A coordinate system in space consists of a fixed point O (the origin) and three directed lines through O that are perpendicular to each other, called the coordinate axes and labeled the x -axis, y -axis, and z -axis. The coordinates of a point $P(a, b, c)$ determine its location relative to the coordinate axes.



The **distance** between the points $P(x_1, y_1, z_1)$ and $Q(x_2, y_2, z_2)$ is given by the

Distance Formula:

$$d(P, Q) = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$



EXAMPLE 10 Using the Distance Formula

$$x_1, y_1, z_1 \quad x_2, y_2, z_2$$

Find the distance between the points $P(2, -1, 7)$ and $Q(1, -3, 5)$.

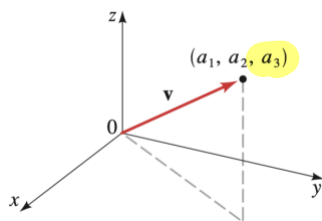
SOLUTION

$$\begin{aligned} d(P, Q) &= \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2} \\ &= \sqrt{(1 - 2)^2 + (-3 - (-1))^2 + (5 - 7)^2} \\ &= \sqrt{1 + 4 + 4} = \sqrt{9} = 3 \end{aligned}$$

Vectors in Three Dimensions

A vector in space is a line segment with a direction. We sketch a vector as an arrow to indicate the direction. A vector in the three dimensional coordinate system is expressed in terms of three coordinates or components

$$\mathbf{v} = \langle a_1, a_2, a_3 \rangle$$



$$\vec{PQ} \neq \vec{OP} - \vec{OQ}$$

If a vector $\mathbf{v}$ has its initial point at $P(x_1, y_1, z_1)$ and its terminal point at $Q(x_2, y_2, z_2)$, then

$$\mathbf{v} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle = \vec{PQ} = \vec{OQ} - \vec{OP}$$

Let $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$, $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$, and $c \in \mathbb{R}$. The operations of vector addition, vector subtraction, scalar multiplication are defined as follows:

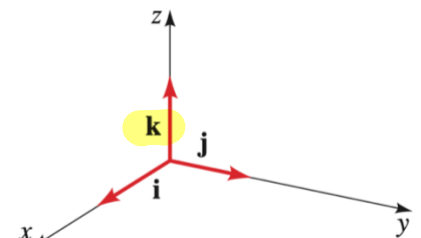
$$\mathbf{u} + \mathbf{v} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

$$\mathbf{u} - \mathbf{v} = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

$$c\mathbf{u} = \langle ca_1, ca_2, ca_3 \rangle$$

The vectors $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are defined by

$$\mathbf{i} = \langle 1, 0, 0 \rangle, \mathbf{j} = \langle 0, 1, 0 \rangle, \mathbf{k} = \langle 0, 0, 1 \rangle$$

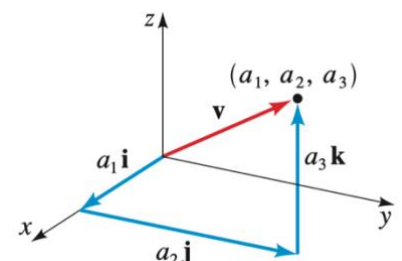


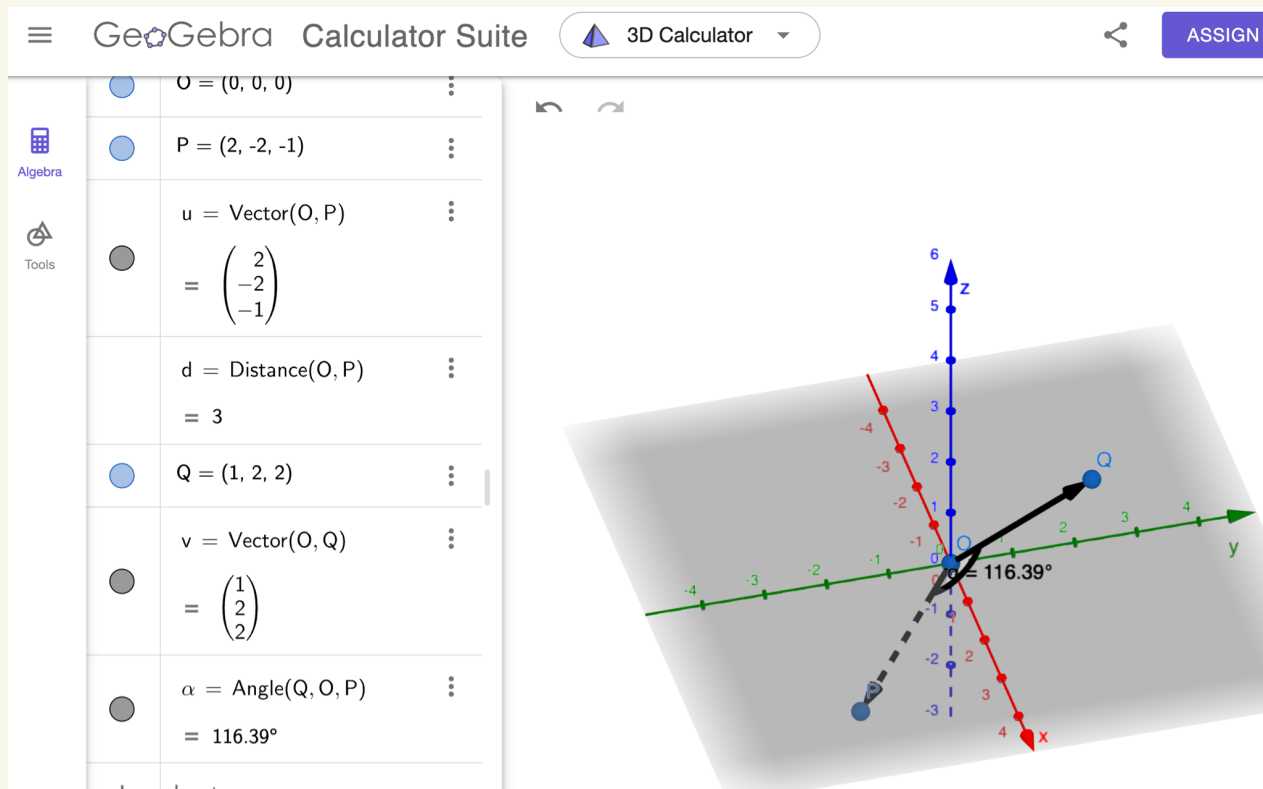
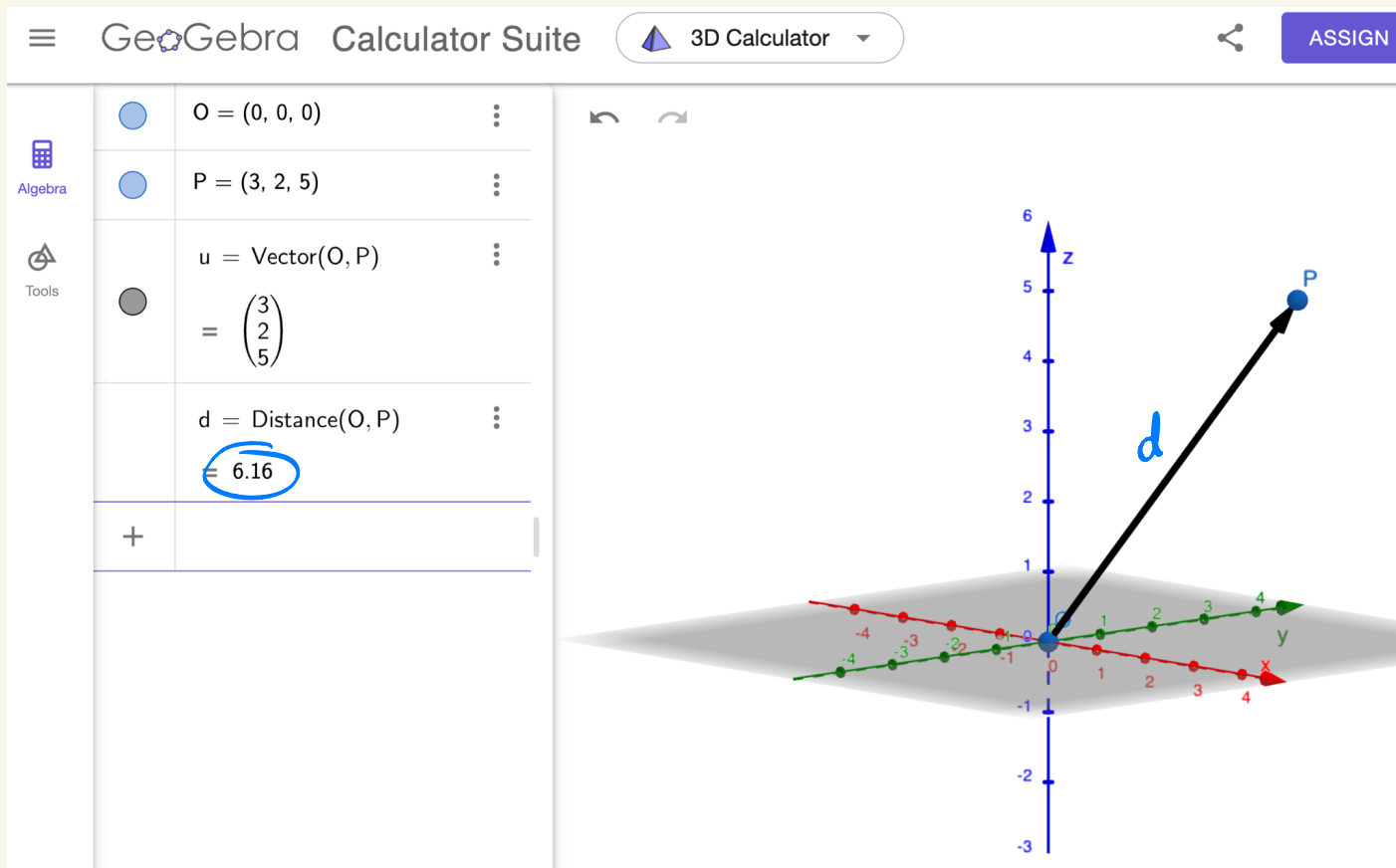
Any vector $\mathbf{v} = \langle a_1, a_2, a_3 \rangle$ can be expressed as

$$\mathbf{v} = a_1\mathbf{i} + a_2\mathbf{j} + a_3\mathbf{k}$$

Let $\mathbf{v} = \langle a_1, a_2, a_3 \rangle$. The magnitude (or length) of $\mathbf{v}$ is

$$|\mathbf{v}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$





$$|\mathbf{v}| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

EXAMPLE 11 Magnitude of Vectors in Three Dimensions

Find the magnitude of the given vector.

- (a) $\mathbf{u} = \langle 3, 2, 5 \rangle$
- (b) $\mathbf{v} = \langle 0, 3, -1 \rangle$
- (c) $\mathbf{w} = \langle 0, 0, -1 \rangle$

SOLUTION

$$\begin{aligned}
 |\mathbf{u}| &= \sqrt{3^2 + 2^2 + 5^2} = \sqrt{9 + 4 + 25} = \sqrt{38} \approx 6.16 \\
 |\mathbf{v}| &= \sqrt{0^2 + 3^2 + (-1)^2} = \sqrt{9 + 1} = \sqrt{10} \approx \\
 |\mathbf{w}| &= \sqrt{0^2 + 0^2 + (-1)^2} = \sqrt{1} = 1 \quad \text{w is a unit vector}
 \end{aligned}$$

EXAMPLE 12 Operations with Three-Dimensional Vectors

If $\mathbf{u} = \langle 1, -2, 4 \rangle$ and $\mathbf{v} = \langle 6, -1, 1 \rangle$ find $\mathbf{u} + \mathbf{v}$, $\mathbf{u} - \mathbf{v}$, and $5\mathbf{u} - 3\mathbf{v}$.

SOLUTION Using the definitions of algebraic operations, we have

$$\begin{aligned}
 \mathbf{u} + \mathbf{v} &= \langle 1 + 6, -2 - 1, 4 + 1 \rangle = \langle 7, -3, 5 \rangle \\
 \mathbf{u} - \mathbf{v} &= \langle 1 - 6, -2 - (-1), 4 - 1 \rangle = \langle -5, -1, 3 \rangle \\
 5\mathbf{u} - 3\mathbf{v} &= 5\langle 1, -2, 4 \rangle - 3\langle 6, -1, 1 \rangle = \langle 5, -10, 20 \rangle - \langle 18, -3, 3 \rangle = \langle -13, -7, 17 \rangle
 \end{aligned}$$

EXAMPLE 13 - Vectors in Terms of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$

- (a) Write the vector $\mathbf{u} = \langle 5, -3, 6 \rangle$ in terms of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.
- (b) If $\mathbf{u} = 2\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}$ and $\mathbf{v} = 4\mathbf{i} + 7\mathbf{k}$, express the vector $2\mathbf{u} + 3\mathbf{v}$ in terms of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$.

SOLUTION

(a) $\mathbf{u} = 5\mathbf{i} + (-3)\mathbf{j} + 6\mathbf{k} = 5\mathbf{i} - 3\mathbf{j} + 6\mathbf{k}$

(b) We use the properties of vectors to get the following:

$$\begin{aligned}
 2\mathbf{u} + 3\mathbf{v} &= 2(2\mathbf{i} + 2\mathbf{j} - 3\mathbf{k}) + 3(4\mathbf{i} + 7\mathbf{k}) \\
 &= 4\mathbf{i} + 4\mathbf{j} - 6\mathbf{k} + 12\mathbf{i} + 21\mathbf{k} \\
 &= 16\mathbf{i} + 4\mathbf{j} + 15\mathbf{k}
 \end{aligned}$$

The Dot Product of Vectors in Space

If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$ are vectors in space, then their dot product is

$$\mathbf{u} \cdot \mathbf{v} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

If θ is the angle between $\mathbf{u}$ and $\mathbf{v}$, then

$$\mathbf{u} \cdot \mathbf{v} = |\mathbf{u}| |\mathbf{v}| \cos \theta$$

The angle θ between $\mathbf{u}$ and $\mathbf{v}$ satisfies

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|}$$

$$\theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|} \right)$$

The vectors $\mathbf{u}$ and $\mathbf{v}$ are perpendicular if and only if



$$\mathbf{u} \cdot \mathbf{v} = 0$$

EXAMPLE 14

Calculating Dot Products for Vectors in Three Dimensions

Find the given dot product.

(a) $\langle -1, 2, 3 \rangle \cdot \langle 6, 5, -1 \rangle$

(b) $(2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) \cdot (-\mathbf{i} + 2\mathbf{j} + 8\mathbf{k})$

SOLUTION

(a) $\langle -1, 2, 3 \rangle \cdot \langle 6, 5, -1 \rangle = (-1)(6) + (2)(5) + (3)(-1) = 1$

(b) $(2\mathbf{i} - 3\mathbf{j} - \mathbf{k}) \cdot (-\mathbf{i} + 2\mathbf{j} + 8\mathbf{k}) = \langle 2, -3, -1 \rangle \cdot \langle -1, 2, 8 \rangle$

$$= (2)(-1) + (-3)(2) + (-1)(8) = -16$$

EXAMPLE 15

Find the angle between $\mathbf{u} = \langle 2, -2, -1 \rangle$ and $\mathbf{v} = \langle 1, 2, 2 \rangle$, rounded to the nearest tenth degree.

SOLUTION

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|} = \frac{(2)(1) + (-2)(2) + (-1)(2)}{\sqrt{2^2 + (-2)^2 + (-1)^2} \sqrt{1^2 + 2^2 + 2^2}} = -\frac{4}{9}$$

$$\theta = \cos^{-1} \left(-\frac{4}{9} \right) \approx 116.39$$

The Cross Product of Vectors in Space

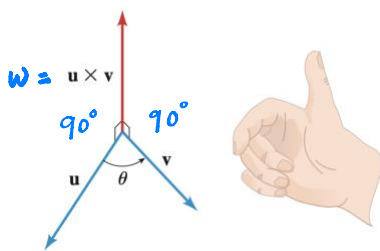
If $\mathbf{u} = \langle a_1, a_2, a_3 \rangle$ and $\mathbf{v} = \langle b_1, b_2, b_3 \rangle$ are vectors in space, then their cross product is the vector

$$\mathbf{u} \times \mathbf{v} = (a_2b_3 - a_3b_2)\mathbf{i} - (a_1b_3 - a_3b_1)\mathbf{j} + (a_1b_2 - a_2b_1)\mathbf{k} \quad \text{Design}$$

We can calculate the cross product using determinants.

$$\mathbf{u} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

The vector $\mathbf{u} \times \mathbf{v}$ is **orthogonal** (or **perpendicular**) to both $\mathbf{u}$ and $\mathbf{v}$.

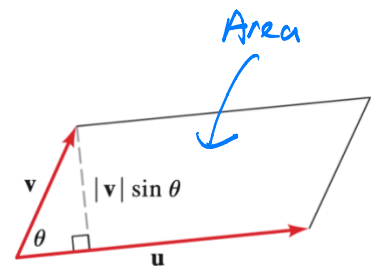


The cross product satisfies

$$|\mathbf{u} \times \mathbf{v}| = |\mathbf{u}||\mathbf{v}|\sin \theta$$

The vectors $\mathbf{u}$ and $\mathbf{v}$ are **parallel** (or **collinear**) if and only if

$$\mathbf{u} \times \mathbf{v} = \mathbf{0}$$



The area of the parallelogram determined by the vectors $\mathbf{u}$ and $\mathbf{v}$ is

$$A = |\mathbf{u} \times \mathbf{v}|$$

The volume of the parallelepiped determined by the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ is

$$V = |\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})|$$

The product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$ is called the **scalar triple product** of the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$. You can check that the scalar triple product can be written as the following determinant:

$$\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix} \quad \checkmark$$

In particular, if the volume of the parallelepiped is 0, then the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are **coplanar**, that is, they lie in the same plane.

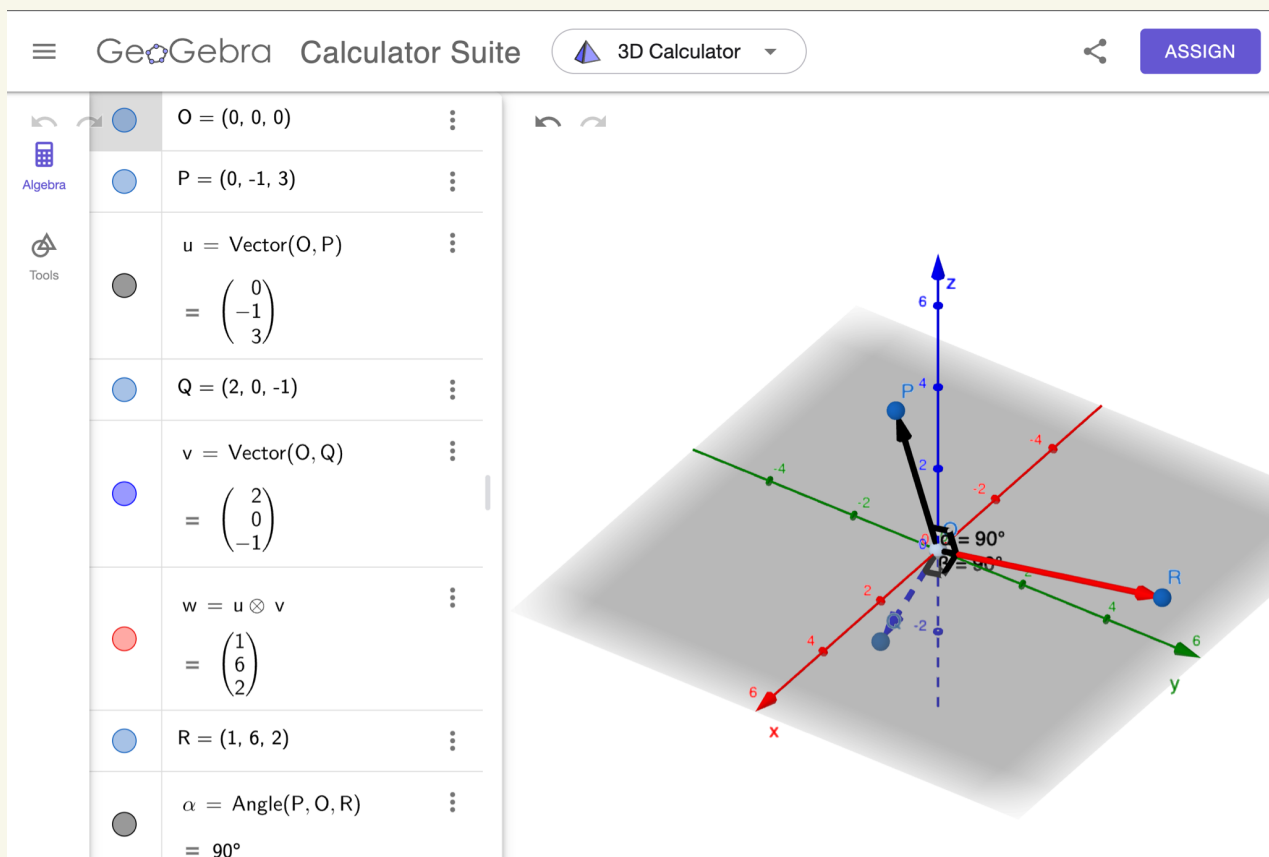
on the same plane.

$$E16 \quad u = \langle 0, -1, 3 \rangle, \quad v = \langle 2, 0, -1 \rangle$$

$$\begin{aligned}
 u \times v &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & 3 \\ 2 & 0 & -1 \end{vmatrix} \\
 &= \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & 3 \\ 2 & 0 & -1 \end{vmatrix} - \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & 3 \\ 2 & 0 & -1 \end{vmatrix} + \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 0 & -1 & 3 \\ 2 & 0 & -1 \end{vmatrix} \\
 &= \begin{vmatrix} -1 & 3 \\ 0 & -1 \end{vmatrix} \hat{i} - \begin{vmatrix} 0 & 3 \\ 2 & -1 \end{vmatrix} \hat{j} + \begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} \hat{k} \\
 &= [(-1)(-1) - (3)(0)] \hat{i} - [0(-1) - (3)(2)] \hat{j} + [0(0) - (-1)(2)] \hat{k} \\
 &= (1-0) \hat{i} - (0-6) \hat{j} + (0-(-2)) \hat{k}
 \end{aligned}$$

$$w = u \times v = \hat{i} + 6\hat{j} + 2\hat{k}$$

$$|u \times v| = |u||v| \sin \theta$$



EXAMPLE 16 Finding a Cross Product

If $\mathbf{u} = \langle 0, -1, 3 \rangle$ and $\mathbf{v} = \langle 2, 0, -1 \rangle$, find $\mathbf{u} \times \mathbf{v}$.

SOLUTION We use the formula above to find the cross product of $\mathbf{u}$ and $\mathbf{v}$:

$$\begin{aligned}\mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -1 & 3 \\ 2 & 0 & -1 \end{vmatrix} \\ &= \begin{vmatrix} -1 & 3 \\ 0 & -1 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & 3 \\ 2 & -1 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & -1 \\ 2 & 0 \end{vmatrix} \mathbf{k} \\ &= (1 - 0)\mathbf{i} - (0 - 6)\mathbf{j} + (0 - (-2))\mathbf{k} \\ &= \mathbf{i} + 6\mathbf{j} + 2\mathbf{k}\end{aligned}$$

So the desired vector is $\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$.

Note that $\mathbf{u}$ and $\mathbf{v}$ are not collinear.

$$\langle -2, 3, 4 \rangle = -12 \langle \frac{1}{6}, -\frac{1}{4}, -\frac{1}{3} \rangle$$

$$\mathbf{u} = -12 \mathbf{v}$$

EXAMPLE 17 Collinear vectors

If $\mathbf{u} = \langle -2, 3, 4 \rangle$ and $\mathbf{v} = \langle \frac{1}{6}, -\frac{1}{4}, -\frac{1}{3} \rangle$, determine whether the vectors $\mathbf{u}$ and $\mathbf{v}$ are collinear.

SOLUTION

u and v are collinear.

$$\begin{aligned}\mathbf{u} \times \mathbf{v} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -2 & 3 & 4 \\ \frac{1}{6} & -\frac{1}{4} & -\frac{1}{3} \end{vmatrix} \\ &= \begin{vmatrix} 3 & 4 \\ -\frac{1}{4} & -\frac{1}{3} \end{vmatrix} \mathbf{i} - \begin{vmatrix} -2 & 4 \\ \frac{1}{6} & -\frac{1}{3} \end{vmatrix} \mathbf{j} + \begin{vmatrix} -2 & 3 \\ \frac{1}{6} & -\frac{1}{4} \end{vmatrix} \mathbf{k} \\ &= \left[3\left(-\frac{1}{3}\right) - 4\left(-\frac{1}{4}\right) \right] \mathbf{i} - \left[(-2)\left(-\frac{1}{3}\right) - 4\left(\frac{1}{6}\right) \right] \mathbf{j} \\ &\quad + \left[(-2)\left(-\frac{1}{4}\right) - 3\left(\frac{1}{6}\right) \right] \mathbf{k} = \mathbf{0}\end{aligned}$$

EXAMPLE 18 Finding an Orthogonal Vector

If $\mathbf{u} = -\mathbf{j} + 3\mathbf{k}$ and $\mathbf{v} = 2\mathbf{i} - \mathbf{k}$, find a unit vector that is orthogonal to the plane containing the vectors $\mathbf{u}$ and $\mathbf{v}$.

$$\mathbf{u} = \langle 0, -1, 3 \rangle \quad \mathbf{v} = \langle 2, 0, -1 \rangle$$

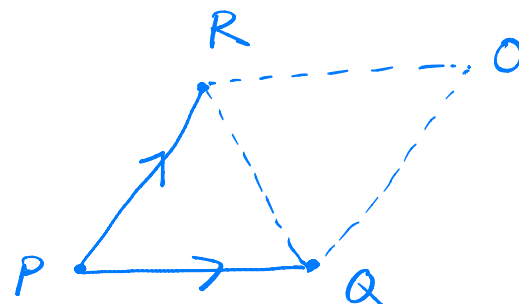
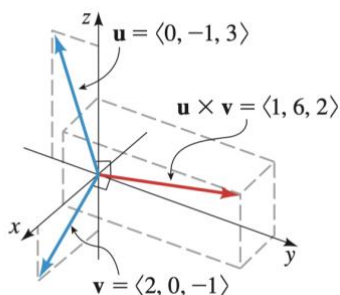
E16

$$\mathbf{w} = \mathbf{u} \times \mathbf{v} = \mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$$

SOLUTION By the Cross Product Theorem the vector $\mathbf{u} \times \mathbf{v}$ is orthogonal to the plane containing the vectors $\mathbf{u}$ and $\mathbf{v}$. In Example 16 we found $\mathbf{u} \times \mathbf{v} = \mathbf{i} + 6\mathbf{j} + 2\mathbf{k}$. To obtain an orthogonal unit vector, we multiply $\mathbf{u} \times \mathbf{v}$ by the scalar $1/|\mathbf{u} \times \mathbf{v}|$:

$$\frac{\mathbf{u} \times \mathbf{v}}{|\mathbf{u} \times \mathbf{v}|} = \frac{\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}}{\sqrt{1^2 + 6^2 + 2^2}} = \frac{\mathbf{i} + 6\mathbf{j} + 2\mathbf{k}}{\sqrt{41}}$$

So the desired vector is $\frac{1}{\sqrt{41}}(\mathbf{i} + 6\mathbf{j} + 2\mathbf{k})$.



EXAMPLE 19 Finding the Area of a Triangle

Find the area of the triangle with vertices $P(1,4,6)$, $Q(-2,5,-1)$, and $R(1,-1,1)$.

SOLUTION

$$\vec{OQ} - \vec{OP}$$

$$\vec{PQ} = (-2 - 1)\mathbf{i} + (5 - 4)\mathbf{j} + (-1 - 6)\mathbf{k} = -3\mathbf{i} + \mathbf{j} - 7\mathbf{k}$$

$$\vec{PR} = (1 - 1)\mathbf{i} + (-1 - 4)\mathbf{j} + (1 - 6)\mathbf{k} = -5\mathbf{j} - 5\mathbf{k}$$

We compute the cross product of these vectors:

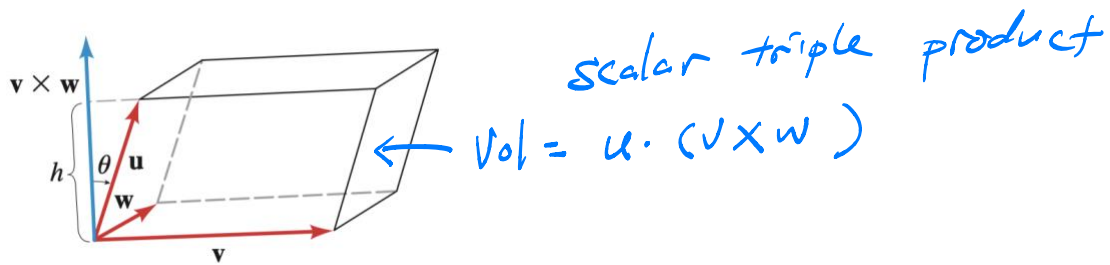
$$\begin{aligned} \vec{PQ} \times \vec{PR} &= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix} \\ &= (-5 - 35)\mathbf{i} - (15 - 0)\mathbf{j} + (15 - 0)\mathbf{k} = -40\mathbf{i} - 15\mathbf{j} + 15\mathbf{k} \end{aligned}$$

$\vec{PQ} \times \vec{PR} = \langle -40, -15, 15 \rangle$. The area of the parallelogram with adjacent sides PQ and PR is the length of this cross product:

$$|\vec{PQ} \times \vec{PR}| = \sqrt{(-40)^2 + (-15)^2 + 15^2} = 5\sqrt{82}$$

The area A of the triangle PQR is half the area of this parallelogram, that is, $\frac{5}{2}\sqrt{82}$.

$$\frac{1}{2} |\vec{PQ} \times \vec{PR}| \rightarrow$$



EXAMPLE 20 Coplanar Vectors

Use the scalar triple product to show that the vectors $\mathbf{u} = \langle 1, 4, -7 \rangle$, $\mathbf{v} = \langle 2, -1, 4 \rangle$, and $\mathbf{w} = \langle 0, -9, 18 \rangle$ are coplanar.

SOLUTION

We compute the scalar triple product:

$$\begin{aligned} \underline{\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})} &= \begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix} \\ &= 1 \begin{vmatrix} -1 & 4 \\ -9 & 18 \end{vmatrix} - 4 \begin{vmatrix} 2 & 4 \\ 0 & 18 \end{vmatrix} + (-7) \begin{vmatrix} 2 & -1 \\ 0 & -9 \end{vmatrix} \\ &= 1(18) - 4(36) - 7(-18) = 0 \end{aligned}$$

So the volume of the parallelepiped is 0, and hence the vectors $\mathbf{u}$, $\mathbf{v}$, and $\mathbf{w}$ are coplanar.

EXAMPLE 21 Volume of the parallelepiped

Let $\mathbf{u} = \langle 2, 3, -2 \rangle$, $\mathbf{v} = \langle -1, 4, 0 \rangle$, $\mathbf{w} = \langle 3, -1, 3 \rangle$.

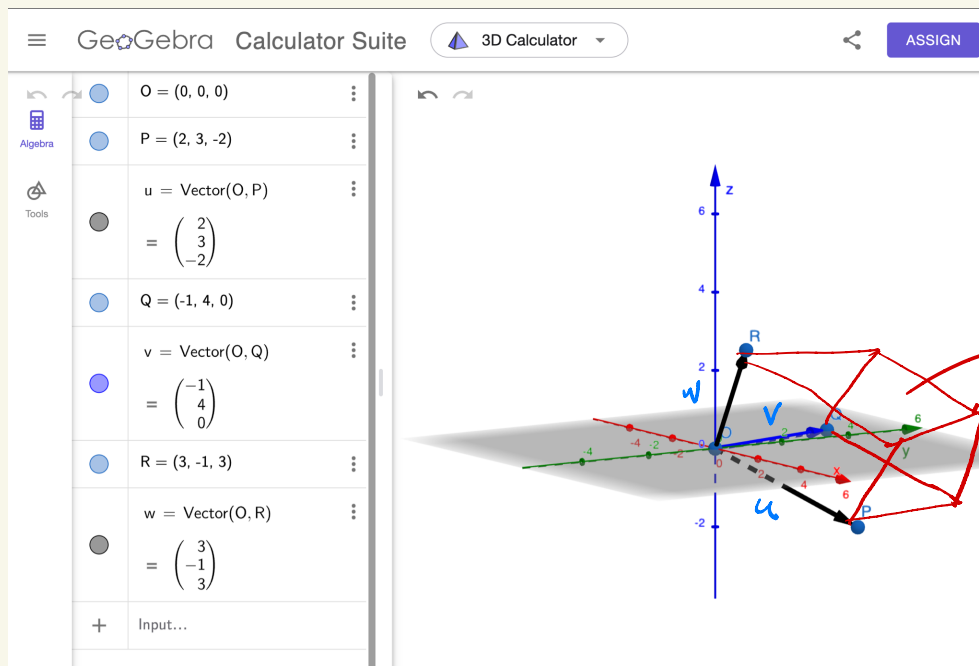
(a) Find their scalar triple product $\mathbf{u} \cdot (\mathbf{v} \times \mathbf{w})$.

(b) Are the vectors coplanar? If not, find the volume of the parallelepiped that they determine.

SOLUTION

$$\begin{aligned} \text{(a)} \quad \mathbf{u} \cdot (\mathbf{v} \times \mathbf{w}) &= \begin{vmatrix} 2 & 3 & -2 \\ -1 & 4 & 0 \\ 3 & -1 & 3 \end{vmatrix} \\ &= \begin{vmatrix} 2 & 3 & -2 \\ -1 & 4 & 0 \\ 3 & -1 & 3 \end{vmatrix} - \begin{vmatrix} 2 & 3 & -2 \\ -1 & 4 & 0 \\ 3 & -1 & 3 \end{vmatrix} + \begin{vmatrix} 2 & 3 & -2 \\ -1 & 4 & 0 \\ 3 & -1 & 3 \end{vmatrix} \\ &= 2 \begin{vmatrix} 4 & 0 \\ -1 & 3 \end{vmatrix} - 3 \begin{vmatrix} -1 & 0 \\ 3 & 3 \end{vmatrix} + (-2) \begin{vmatrix} -1 & 4 \\ 3 & -1 \end{vmatrix} \\ &= 2[4(3) - 0(-1)] - 3[(-1)(3) - 0(3)] + (-2)[(-1)(-1) - (4)(3)] \\ &= 2(12) - 3(-3) + (-2)(-11) = 55 \neq 0 \end{aligned}$$

(b) The vectors are not coplanar.
Vol = 55.



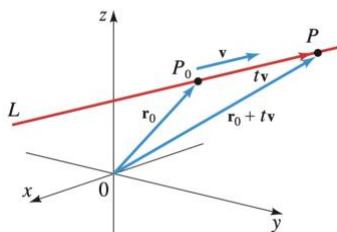
$V_0 = 55$

Equations of Lines and Planes

A line L in three-dimensional space is determined when we know a point $P_0(x_0, y_0, z_0)$ on L and the direction of L . In three dimensions the direction of a line is described by a vector $\mathbf{v}$ parallel to L . If we let $\mathbf{r}_0$ be the position vector of P_0 (that is, the vector $\overrightarrow{OP_0}$), then for all real numbers t the terminal points P of the position vectors $\mathbf{r}_0 + t\mathbf{v}$ trace out a line parallel to $\mathbf{v}$ and passing through P_0 . Each value of the parameter t gives a point P on L . So the line L is given by the position vector $\mathbf{r}$, where

$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$$

for $t \in \mathbb{R}$. This is the vector equation of a line.



$$\mathbf{r} = \mathbf{r}_0 + t\mathbf{v}$$

Let's write the vector $\mathbf{v}$ in component form $\mathbf{v} = \langle a, b, c \rangle$ and let $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$ and $\mathbf{r} = \langle x, y, z \rangle$. Then the vector equation of the line becomes

$$\begin{aligned} \mathbf{r} = \langle x, y, z \rangle &= \mathbf{r}_0 + t\mathbf{v} \\ &= \langle x_0 + ta, y_0 + tb, z_0 + tc \rangle \end{aligned} \quad \text{vector equation}$$

Since two vectors are equal if and only if their corresponding components are equal, we have the following result.

A line passing through the point $P(x_0, y_0, z_0)$ and parallel to the vector $\mathbf{v} = \langle a, b, c \rangle$ is described by the parametric equations

$$\begin{cases} x = x_0 + at \\ y = y_0 + bt \\ z = z_0 + ct \end{cases}$$

t is a parameter

where t is any real number.

EXAMPLE 22 Equations of a Line

$$(x_0, y_0, z_0)$$

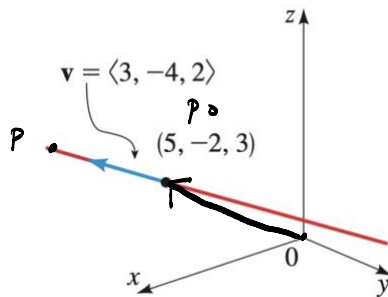
Find parametric equations for the line that passes through the point $(5, -2, 3)$ and is parallel to the vector $\mathbf{v} = \langle 3, -4, 2 \rangle$.

SOLUTION We use the above formula to find the parametric equations:

$$\langle a, b, c \rangle$$

$$\begin{aligned}x &= 5 + 3t \\y &= -2 - 4t \\z &= 3 + 2t\end{aligned}$$

where t is any real number.



EXAMPLE 23 Equations of a Line

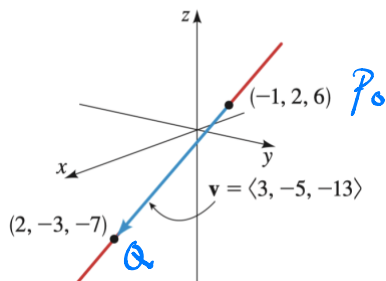
Find parametric equations for the line that passes through the points $\underline{(-1, 2, 6)}$ and $\underline{(2, -3, -7)}$. P_0 Q

SOLUTION

$$\begin{aligned}r_0 &= \langle x_0, y_0, z_0 \rangle = \vec{OP}_0 \\r_0 &= \langle -1, 2, 6 \rangle \\v &= \vec{P_0Q} = \vec{OQ} - \vec{OP}_0 \\v &= \langle 2, -3, -7 \rangle - \langle -1, 2, 6 \rangle \\v &= \langle 3, -5, -13 \rangle \\v &= \langle a, b, c \rangle\end{aligned}$$

$$\begin{cases}x = x_0 + at \\y = y_0 + bt \\z = z_0 + ct\end{cases}$$

$$\begin{cases}x = -1 + 3t \\y = 2 - 5t \\z = 6 - 13t \\t \in \mathbb{R}\end{cases}$$



A plane in space is determined by a point $P_0(x_0, y_0, z_0)$ in the plane and a vector $\mathbf{n}$ that is orthogonal to the plane. This orthogonal vector $\mathbf{n}$ is called a normal vector. To determine

whether a point $P(x, y, z)$ is in the plane, we check whether the vector $\overrightarrow{P_0P}$ with initial point P_0 and terminal point P is orthogonal to the normal vector. Let $\mathbf{r}_0$ and $\mathbf{r}$ be the position vectors of P_0 and P , respectively. Then the vector $\overrightarrow{P_0P}$ is represented by $\mathbf{r} - \mathbf{r}_0$. So the plane is described by the tips of the vectors $\mathbf{r}$ satisfying

$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$$

This is the **vector equation of the plane**.



Let's write the normal vector $\mathbf{n}$ in component form $\mathbf{n} = \langle a, b, c \rangle$ and let $\mathbf{r}_0 = \langle x_0, y_0, z_0 \rangle$ and $\mathbf{r} = \langle x, y, z \rangle$. Then the vector equation of the plane becomes

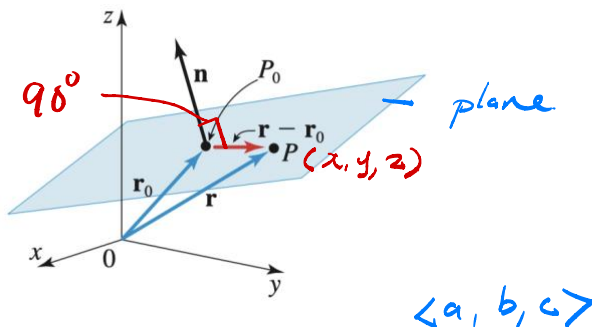
$$\mathbf{n} \cdot (\mathbf{r} - \mathbf{r}_0) = 0$$

$$\langle a, b, c \rangle \cdot \langle x - x_0, y - y_0, z - z_0 \rangle = 0$$

Performing the dot product, we arrive at the following equation of the plane in the variables x, y , and z .

A plane containing the point $P(x_0, y_0, z_0)$ and having the normal vector $\mathbf{n} = \langle a, b, c \rangle$ is described by the **plane equation**

$$a(x - x_0) + b(y - y_0) + c(z - z_0) = 0$$



EXAMPLE 24 Finding an Equation for a Plane

A plane has normal vector $\mathbf{n} = \langle 4, -6, 3 \rangle$ and passes through the point $P(3, -1, -2)$.

- Find an equation of the plane.
- Find the intercepts, and sketch a graph of the plane.

SOLUTION

(a) By the above formula for the equation of a plane we have

$$4(x - 3) - 6(y - (-1)) + 3(z - (-2)) = 0 \quad \text{Formula}$$

$$4x - 12 - 6y - 6 + 3z + 6 = 0 \quad \text{Expand}$$

$$4x - 6y + 3z = 12 \quad \text{Simplify}$$

$$\text{or } 4x - 6y + 3z - 12 = 0$$

Thus an equation of the plane is $4x - 6y + 3z = 12$.

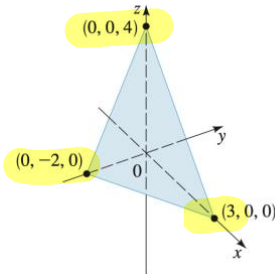
(b) To find the x -intercept, we set $y = 0$ and $z = 0$ in the equation of the plane and solve for x . Similarly, we find the y - and z -intercepts.

x-intercept: Setting $y = 0, z = 0$, we get $x = 3$.

y-intercept: Setting $x = 0, z = 0$, we get $y = -2$.

z-intercept: Setting $x = 0, y = 0$, we get $z = 4$.

So the graph of the plane intersects the coordinate axes at the points $(3,0,0)$, $(0,-2,0)$, and $(0,0,4)$.



EXAMPLE 25 Finding an Equation for a Plane

$P(x_0, y_0, z_0)$

Find an equation of the plane that passes through the points $P(1,4,6)$, $Q(-2,5,-1)$, and $R(1,-1,1)$.

SOLUTION

From **E19**

$$\vec{PQ} = -3\vec{i} + \vec{j} - 7\vec{k}$$

$$\vec{PR} = -5\vec{j} - 5\vec{k}$$

$$n = \vec{PQ} \times \vec{PR}$$

$$= -40\vec{i} - 15\vec{j} + 15\vec{k}$$

$$= \langle -40, -15, 15 \rangle$$

$$= \langle a, b, c \rangle$$

$$a(x-x_0) + b(y-y_0) + c(z-z_0) = 0$$

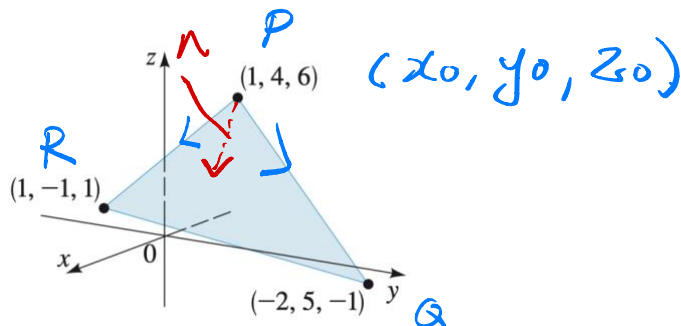
$$-40(x-1) - 15(y-4) + 15(z-6) = 0$$

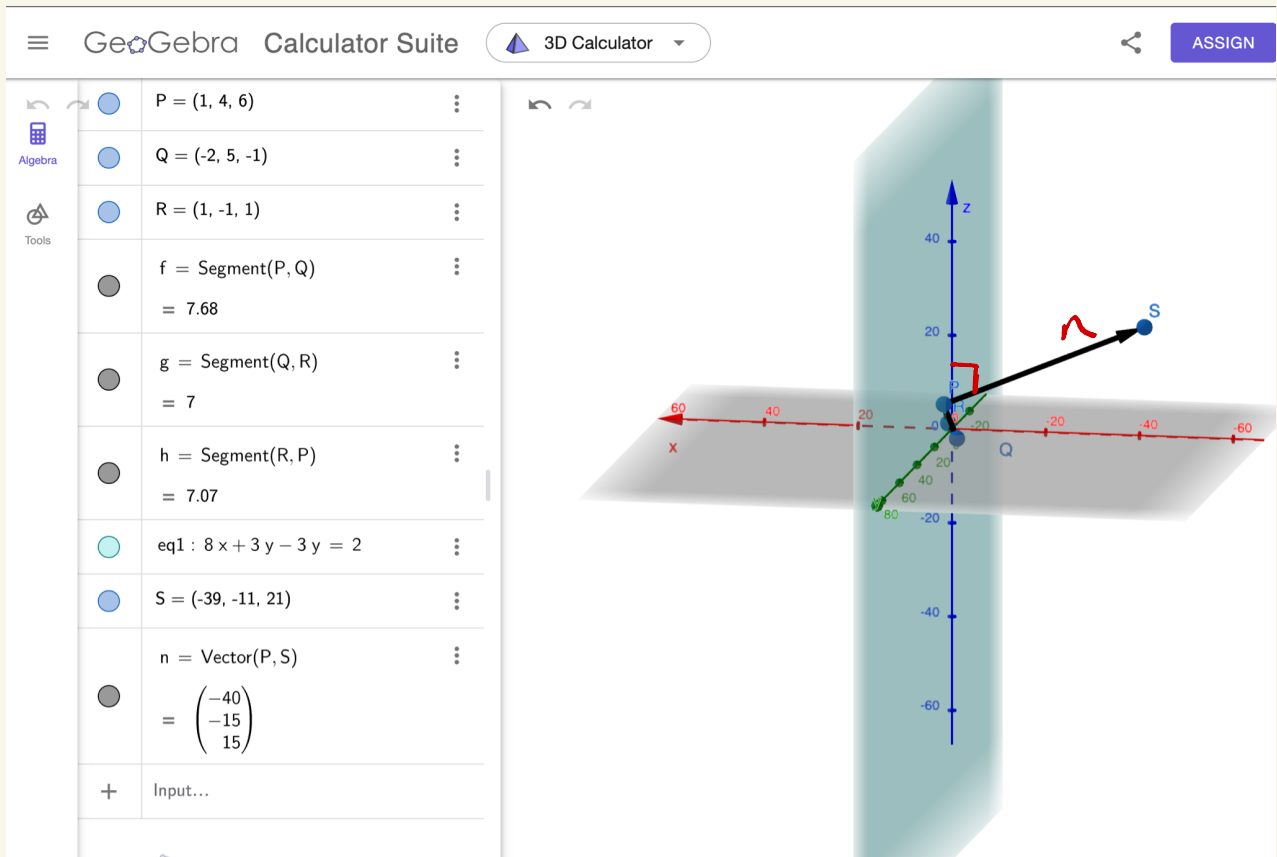
$$-40x + 40 - 15y + 60 + 15z - 90 = 0$$

$$-40x - 15y + 15z = -10$$

$$8x + 3y - 3z = 2$$

$$8x + 3y - 3z - 2 = 0$$





E25 E19

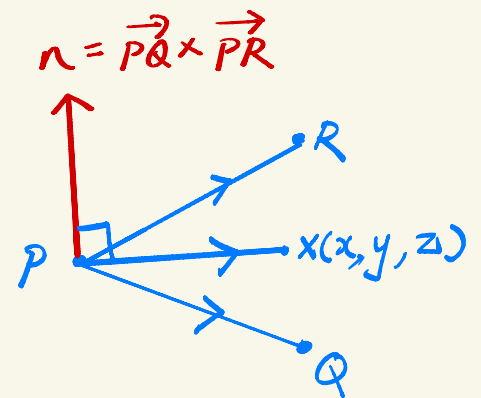
$$P(1, 4, 6), Q(-2, 5, -1), R(1, -1, 1)$$

$$\begin{aligned}\vec{PQ} &= \vec{OQ} - \vec{OP} \\ &= \langle -2-1, 5-4, -1-6 \rangle \\ &= \langle -3, 1, -7 \rangle\end{aligned}$$

$$\begin{aligned}\vec{PR} &= \vec{OR} - \vec{OP} \\ &= \langle 1-1, -1-4, 1-6 \rangle \\ &= \langle 0, -5, -5 \rangle\end{aligned}$$

$$n = \vec{PQ} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix}$$

$$= \begin{vmatrix} \textcircled{\vec{i}} & \cancel{\vec{j}} & \cancel{\vec{k}} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix} - \begin{vmatrix} \cancel{\vec{i}} & \textcircled{\vec{j}} & \cancel{\vec{k}} \\ -3 & \cancel{1} & -7 \\ 0 & -5 & -5 \end{vmatrix} + \begin{vmatrix} \cancel{\vec{i}} & \cancel{\vec{j}} & \textcircled{\vec{k}} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix}$$



$$= \begin{vmatrix} 1 & -7 \\ -5 & -5 \end{vmatrix} \hat{i} - \begin{vmatrix} -3 & -7 \\ 0 & -5 \end{vmatrix} \hat{j} + \begin{vmatrix} -3 & 1 \\ 0 & -5 \end{vmatrix} \hat{k}$$

$$= (-5 - 35)\hat{i} - (15 - 0)\hat{j} + (15 - 0)\hat{k}$$

$$= -40\hat{i} - 15\hat{j} + 15\hat{k}$$

$$n = \langle -40, -15, 15 \rangle$$

$$n = \langle a, b, c \rangle$$

Lines and Planes

A line is **parallel** to the vector **$\mathbf{v}$** , and a plane has normal vector **$\mathbf{n}$** .

(a) If the **line is perpendicular to the plane**, what is the relationship between **$\mathbf{v}$** and **$\mathbf{n}$** (parallel or perpendicular)?

(b) If the **line is parallel to the plane** (that is, the line and the plane do not intersect), what is the relationship between **$\mathbf{v}$** and **$\mathbf{n}$** (parallel or perpendicular)?

(c) Parametric equations for two lines are given.

(a) Line 1: $x = 2t, y = 3 - 2t, z = 4 + 8t$

(b) Line 2: $x = -2t, y = 5 + 2t, z = 3 + t$

Which line is parallel to the plane $x - y + 4z = 6$?

Which line is perpendicular to this plane?

SOLUTION

(a) If the **line is perpendicular to the plane**, then it is parallel to the plane's normal vector, so **$\mathbf{v}$** is parallel to **$\mathbf{n}$** .

(b) If the **line is parallel to the plane**, then it is perpendicular to the plane's normal vector, so **$\mathbf{v}$** is perpendicular to **$\mathbf{n}$** .

(c) The plane's normal vector is **$\mathbf{n} = \langle 1, -1, 4 \rangle$** .

Line 1 is parallel to **$\langle 2, -2, 8 \rangle = 2\langle 1, -1, 4 \rangle$** , so, as in part (a), it is perpendicular to the given plane.

Line 2 is parallel to **$\mathbf{v} = \langle -2, 2, 1 \rangle$** , and **$\mathbf{v} \cdot \mathbf{n} = -2 - 2 + 4 = 0$** , so as in part (b), it is parallel to the given plane.

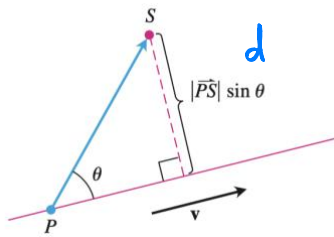
shortest.

The Distance from a Point to a Line in Space

To find the distance from a point S to a line that passes through a point P parallel to a vector **$\mathbf{v}$** , we find the absolute value of the scalar component of $\overrightarrow{PS}$ in the direction of a vector normal to the line

$$\sin \theta = \frac{d}{|\overrightarrow{PS}|}$$

$$d = |\overrightarrow{PS}| \sin \theta$$



$$\frac{|\vec{PS} \times \mathbf{v}|}{|\mathbf{v}|} = \frac{|\vec{PS}| |\mathbf{v}| \sin \theta}{|\mathbf{v}|} = |\vec{PS}| \sin \theta$$

In the notation of the figure, the absolute value of the scalar component is, $|\vec{PS}| \sin \theta$, which is $\frac{|\vec{PS} \times \mathbf{v}|}{|\mathbf{v}|}$.

Distance from a Point S to a Line Through P Parallel to $\mathbf{v}$

$$d = \frac{|\vec{PS} \times \mathbf{v}|}{|\mathbf{v}|}$$

EXAMPLE 26 Finding Distance from a Point to a Line

Find the distance from the point $S(1,1,5)$ to the line

$$L: x = 1 + t, y = 3 - t, z = 2t.$$

Solution We see from the equations for L that L passes through $P(1,3,0)$ parallel to $\mathbf{v} = \mathbf{i} - \mathbf{j} + 2\mathbf{k}$. With $\vec{PS} = \vec{OS} - \vec{OP}$

$$\vec{PS} = (1 - 1)\mathbf{i} + (1 - 3)\mathbf{j} + (5 - 0)\mathbf{k} = -2\mathbf{j} + 5\mathbf{k}$$

and

$$\vec{PS} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = \mathbf{i} + 5\mathbf{j} + 2\mathbf{k},$$

$$d = \frac{|\vec{PS} \times \mathbf{v}|}{|\mathbf{v}|} = \frac{\sqrt{1 + 25 + 4}}{\sqrt{1 + 1 + 4}} = \frac{\sqrt{30}}{\sqrt{6}} = \sqrt{5}.$$

cross

$$\vec{PS} \times \mathbf{v} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} - \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix} + \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ 0 & -2 & 5 \\ 1 & -1 & 2 \end{vmatrix}$$

$$= \begin{vmatrix} -2 & 5 \\ -1 & 2 \end{vmatrix} \mathbf{i} - \begin{vmatrix} 0 & 5 \\ 1 & 2 \end{vmatrix} \mathbf{j} + \begin{vmatrix} 0 & -2 \\ 1 & -1 \end{vmatrix} \mathbf{k}$$

$$= [-4 - (-5)]\mathbf{i} - [0 - 5]\mathbf{j} + [0 - (-2)]\mathbf{k}$$

$$= \underline{\underline{2 + 5j + 2k}}$$

The Distance from a Point to a Plane

If P is a point on a plane with normal $\mathbf{n}$, then the distance from any point S to the plane is the length of the vector projection of $\overrightarrow{PS}$ onto $\mathbf{n}$. That is, the distance from S to the plane is

simple

$$d = \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right|$$

where $\mathbf{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$ is normal to the plane.

$$d = |\overrightarrow{PS}| \cos \theta$$

$$\overrightarrow{PS} \cdot \mathbf{n} = |\overrightarrow{PS}| |\mathbf{n}| \cos \theta$$

dot

$$\left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| = \frac{|\overrightarrow{PS}| |\mathbf{n}| \cos \theta}{|\mathbf{n}|}$$

EXAMPLE 27 Finding the Distance from a Point to a Plane

Find the distance from $S(1,1,3)$ to the plane $3x + 2y + 6z = 6$.

$$x=0 \quad z=0$$

$$\mathbf{n} = \langle 3, 2, 6 \rangle$$

$$P(0, 3, 0)$$

$$2y=6$$

$$y=3$$

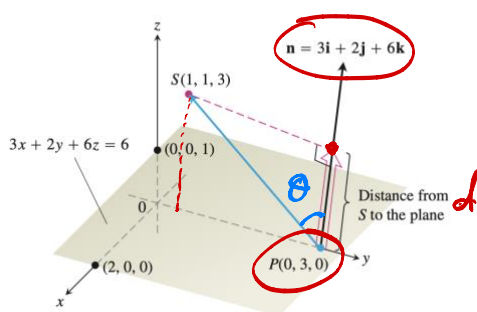
$$= |\overrightarrow{PS}| \cos \theta$$

$$= d$$

SOLUTION

We find a point P in the plane and calculate the length of the vector projection of $\overrightarrow{PS}$ onto a vector $\mathbf{n}$ normal to the plane (Figure 12.41). The coefficients in the equation $3x + 2y + 6z = 6$ give

$$\mathbf{n} = 3\mathbf{i} + 2\mathbf{j} + 6\mathbf{k} \quad \checkmark$$



$$|\cos \theta| = \frac{d}{|\overrightarrow{PS}|}$$

$$d = |\overrightarrow{PS}| |\cos \theta|$$

The points on the plane easiest to find from the plane's equation are the intercepts. If we take P to be the y-intercept $(0, 3, 0)$, then

$$\overrightarrow{PS} = (1 - 0)\mathbf{i} + (1 - 3)\mathbf{j} + (3 - 0)\mathbf{k}$$

$$= \mathbf{i} - 2\mathbf{j} + 3\mathbf{k},$$

$$|\mathbf{n}| = \sqrt{(3)^2 + (2)^2 + (6)^2} = \sqrt{49} = 7.$$

The distance from S to the plane is

$$\overrightarrow{OP} + \overrightarrow{PS} = \overrightarrow{OS}$$

$$\overrightarrow{PS} = \overrightarrow{OS} - \overrightarrow{OP}$$

$$\begin{aligned}
 d &= \left| \overrightarrow{PS} \cdot \frac{\mathbf{n}}{|\mathbf{n}|} \right| \\
 &= \left| (\mathbf{i} - 2\mathbf{j} + 3\mathbf{k}) \cdot \left(\frac{3}{7}\mathbf{i} + \frac{2}{7}\mathbf{j} + \frac{6}{7}\mathbf{k} \right) \right| \\
 &= \left| \frac{3}{7} - \frac{4}{7} + \frac{18}{7} \right| = \frac{17}{7} \approx 2.43
 \end{aligned}$$

